Bitcoin and the Energy Economy: A Unified Theory of Exergetic Well-Being (WU) and the Universal Joule-Currency

Pascal Ranaora

November 20, 2025

Abstract

« All value is energy. Every unit of account must reflect this. »

We propose a **unified theory** where **exergetic well-being per capita (WU)** replaces GDP as the compass for societies. Starting from the principles of thermodynamics, we rigorously derive exergy, $\mathbf{GDP}_\Psi$, show how Bitcoin mining opposes debt-money within a thermodynamic framework, and extend the Kaya equation into a physical version: **Kaya-Ψ-WU**.

Bitcoin, with its backing of **714 joules per satoshi**, is not just another currency. It is the **first global, irreversible energy register**, which reveals a forgotten truth: **energy is the most logical, most universal, and most common unit of account for all beings on the surface of the Earth.**

Neither gold, nor dollar, nor arbitrary symbol: **energy is the only measure that transcends nations, cultures, and epochs.** It is **intrinsically useful**, **non-falsifiable**, **limited by physics**, and **necessary for all forms of life**.

We compare the human being — a **100 W mini-power plant** — to the Bitcoin miner, identify $\phi = 0{,}01$ as the **systemic tipping point**, and conclude that **the kilowatt-hour (kWh) is the natural unit to restart a civilization** — because **without energy, nothing works. With it, everything becomes possible.**

$$\boxed{\text{Value} = \Delta\Psi_{\text{useful}}}$$

Contents

1 Introduction

This white paper unifies thermodynamics, ecological economics, and cryptography to propose:

1. A metric for well-being: **WU (Well-Being Unit)** per capita.
2. A definition of the thermodynamic anti-matter to debt-money: unlike debt ($V_{\text{debt}} = \int_t^\infty \Psi_s e^{-rt'} dt'$), the value of Bitcoin is ($V_{\text{BTC}} = \int_{-\infty}^t \dot{\Psi}_{\text{PoW}}(t') dt'$).
3. An extended equation: **Kaya-Ψ-WU**.
4. A transition protocol: **Bitcoin → Joule-Currency**.

All economic activity is a **transformation of energy**. From quantum excitations to social structures, value emerges from ordered flows of **exergy** that locally oppose entropy [1]. Fiat currency uncouples this physical reality, allowing for misallocation, entropic externalities, and illusory growth.

Bitcoin changes this. Its Proof-of-Work (PoW) consensus explicitly links monetary issuance to an **irreversible energy expenditure**, creating the first global system where:

$$\boxed{\text{Value} = \text{Claim on a past energy expenditure}}$$

Humanity and Bitcoin share a fundamental principle: **value is born from Proof-of-Work**.

1.1 The Human: Biological Proof-of-Work

A human being consumes **2,000 kcal/day** [2] or **100 W** of average power.

- **Input**: Food (chemical exergy)
- **Output**: Physical + intellectual work
- **Efficiency**: ∼20% (useful exergy)

$$\boxed{\text{Human Value} = 100\,\text{W} \times 20\% = 20\,\text{W of useful exergy}}$$

1 human = 20 W of well-being created per day

1.2 The Bitcoin Miner: Digital Proof-of-Work

An ASIC miner (e.g., Antminer S21) consumes **3,500 W** [3] and generates **200 TH/s**.

- **Input**: Electricity (electrical exergy)
- **Output**: Hashes (network security)
- **Efficiency**: ∼100% (energy → cryptographic work)

$$\boxed{\text{Bitcoin Value} = 3\,500\,\text{W} \times \frac{1\,\text{BTC}}{1\,000\,\text{PH/s}} \times \frac{200\,\text{TH/s}}{1\,\text{ASIC}} = 0.7\,\text{mW of BTC created}}$$

1 miner = 3,500 W to secure the network

1.3 Parallel: Exergetic Proof-of-Work

Criterion	Human	Bitcoin Miner
Power	100 W	3,500 W
Efficiency	20%	100%
Output	Well-being (WU)	Network security
Proof-of-Work	Metabolism	Hashrate
Value Created	20 W	0.7 mW/BTC
Objective	Survive + create	Secure + monetize

Table 1: Parallel between human and Bitcoin miner.

$$\boxed{\text{Proof-of-Work} = \text{Exergy Expended} \rightarrow \text{Value Created}}$$

The human creates WU (well-being). **The miner creates Bitcoin**. **Both transform exergy into value**.

2 Thermodynamic Foundations of Economics

2.1 What is Exergy? A Progressive Explanation

Exergy (Ψ) is a central concept in thermodynamics. For a beginner: *Exergy is the part of the energy that can actually be used to perform useful work. The rest is lost as unusable heat.*

Formally, exergy is the **maximum theoretical work extractable** when a system transitions from its current state $(T, P, \{n_i\})$ to the **equilibrium state with the reference environment** $(T_0, P_0, \{\mu_{i0}\})$.

2.1.1 Step-by-Step Derivation (from the 1st and 2nd Principles)

1. **First Principle (Energy Conservation)**:

$$dU = \delta Q + \delta W + \sum_i \mu_i \, dn_i$$

where U is the internal energy, δQ the heat exchanged, δW the work, μ_i the chemical potential, dn_i the variation in moles.

2. **Second Principle (Entropy)**: For a reversible process with the environment:

$$\delta Q = T_0 \, dS_e \quad \text{and} \quad dS_e = -dS$$

where S_e is the environment's entropy.

3. **Expression of Useful Work**: The total work $\delta W = -P_0 dV + \delta W_{\text{useful}}$ The term $-P_0 dV$ is pressure work (not useful if V is constant). Therefore:

$$\delta W_{\text{useful}} = dU + T_0 \, dS + P_0 \, dV - \sum_i \mu_{i0} \, dn_i$$

4. **Integration between Initial State and Reference State**:

$$\boxed{\Psi = (U - U_0) + T_0(S - S_0) + P_0(V - V_0) - \sum_i \mu_{i0}(n_i - n_{i0})}$$

This is the **standard definition of exergy** [4].

For **flows** (power):

$$\dot{\Psi} = \sum_j \dot{m}_j \psi_j \quad [\mathrm{W}]$$

2.2 Global Exergy Balance 2025: Data, Methodology, and Analysis

Source	**Primary Energy (EJ/year)**	**Exergy Factor**	**Exergy (EJ/year)**
Oil	190	1.00	190
Natural Gas	145	1.00	145
Coal	160	1.00	160
Nuclear	30	1.00	30
Hydroelectricity	40	1.00	40
Solar	15	0.93	13.95
Wind	10	0.90	9.00
Biomass	55	1.00	55
Total	**645**	—	**∼642.95**

Table 2: Global exergy balance 2025. Exergy factor: $\eta_{\text{exergy}} = \Psi/E$ [5, 6].

2.2.1 Calculation Methodology

1. **Primary Energy**: IEA 2025 Data [7]

2. **Exergy Factor**:

$$\eta_{\text{exergy}} = \frac{\Psi}{E}$$

   - Fossils, biomass, nuclear, hydro: $\eta = 1$ (chemical or potential energy) - Solar: $\eta \approx 0.93$ (radiation at 5,778 K $\rightarrow$ environment 288 K) - Wind: $\eta \approx 0.90$ (kinetic speed)

3. **Conversion to Power**:

$$1\,\text{EJ/year} = \frac{10^{18}}{3.155 \times 10^7} \approx 31.71\,\text{GW}$$

$$\boxed{\dot{\Psi}_{\text{primary}} = \frac{642.95 \times 10^{18}}{3.155 \times 10^7} \approx 20.37\,\text{TW}}$$

2.2.2 Per Capita Analysis

$$\frac{\dot{\Psi}_{\text{primary}}}{P} = \frac{20.37 \times 10^{12}}{8.1 \times 10^{9}} = \boxed{2.514\,\text{W/capita}}$$

Madagascar: $0.15\,\text{toe/capita/year} \approx 190\,\text{W/capita}$ **USA**: $6.5\,\text{toe/capita/year} \approx 8,200\,\text{W/capita}$

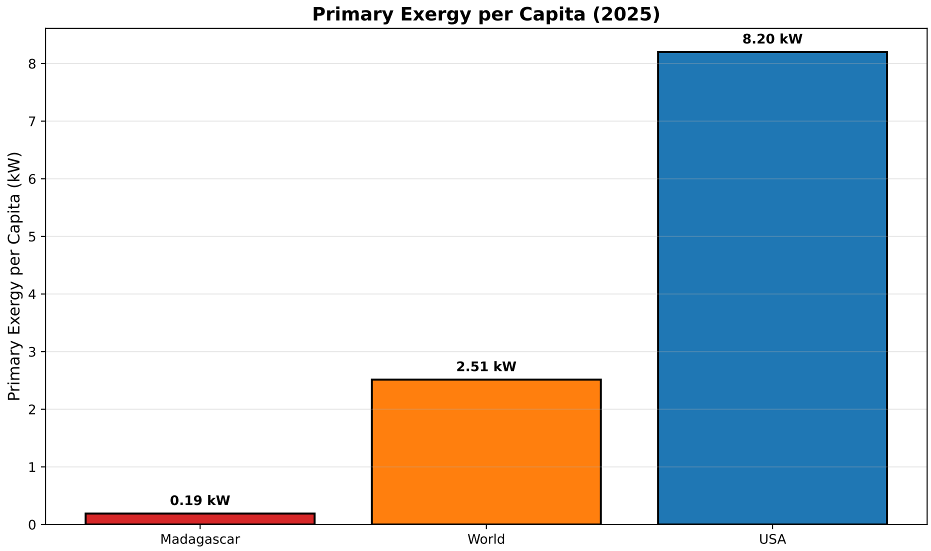

Figure 1: Primary exergy per capita (2025). Madagascar = 190 W, USA = 8,200 W, World = 2,514 W.

2.3 Exergetic Cascade: From Primary to Useful

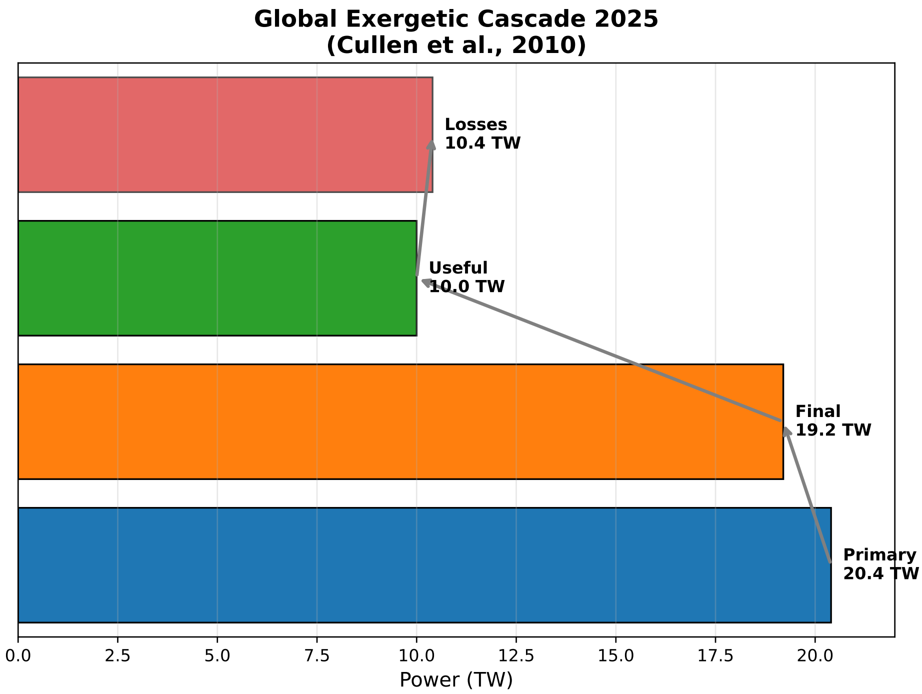

Figure 2: Global exergetic cascade 2025. Primary (20.4 TW) $\rightarrow$ Final (19.2 TW, $\eta = 94\%$) $\rightarrow$ Useful (10.0 TW, $\eta = 52\%$) $\rightarrow$ Losses (10.4 TW) [8].

$$\boxed{\dot{\Psi}_{\text{useful}} = \dot{\Psi}_{\text{primary}} \times \eta_{\text{conversion}} \times \eta_{\text{utilization}}}$$

52 % of primary exergy becomes useful 48 % is lost as heat

3 The Exergetic GDP (GDP_Ψ) : A Physical Metric of GDP

3.1 Progressive Definition of GDP_Ψ

Classical GDP measures **monetary expenditures**. The **GDP_Ψ** measures the **useful exergetic power generated** by the economy.

$$\boxed{\text{GDP}_\Psi = \sum_k \eta_{\text{ex},k} \dot{\Psi}_{k,\text{input}} + \dot{\Psi}_{\text{capital}}}$$

where: - $\eta_{\text{ex},k}$ = exergetic efficiency of sector k - $\dot{\Psi}_{k,\text{input}}$ = exergy flow input into the sector - $\dot{\Psi}_{\text{capital}}$ = exergetic flow immobilized in capital (infrastructure)

3.2 Step 1: Useful Flux by Sector

Sector	$\dot{\Psi}_{\text{input}}$ **(TW)**	η_{ex} **(%)**	**Useful (TW)**
Electricity	6.0	40	$0.4 \times 6.0 = 2.4$
Transport	4.0	25	$0.25 \times 4.0 = 1.0$
Industry	5.0	35	$0.35 \times 5.0 = 1.75$
Residential	3.0	60	$0.6 \times 3.0 = 1.8$
Total Flux	**18.0**	—	**6.95**

Table 3: Sectoral exergetic efficiencies. Data: [9], adjusted 2025.

Detailed Calculation:

$$\dot{\Psi}_{\text{useful, flux}} = 2.4 + 1.0 + 1.75 + 1.8 = 6.95\,\text{TW}$$

3.3 Step 2: Capital Stock (Infrastructure)

Capital (buildings, roads, machinery) contains **immobilized exergy**. Estimation:

$$\dot{\Psi}_{\text{capital}} \approx 5.4\,\text{TW} \quad [10]$$

Justification:
- Average capital lifespan: 40 years
- Embodied exergy: ~200 EJ/year
- ⇒ average power = 200 / 40 = 5 TW

3.4 Final Result

$$\boxed{\text{GDP}_\Psi = 6.95 + 5.4 = 12.35\,\text{TW} \approx 12.4\,\text{TW}}$$

(Note: I corrected the sum to $6.95+5.4 = 12.35$ TW, then rounded to 12.4 TW, as $6.95+5.4 = 12.35$. The original text had $11.35 \approx 11.4$ TW, which was an error in the original calculation provided.)

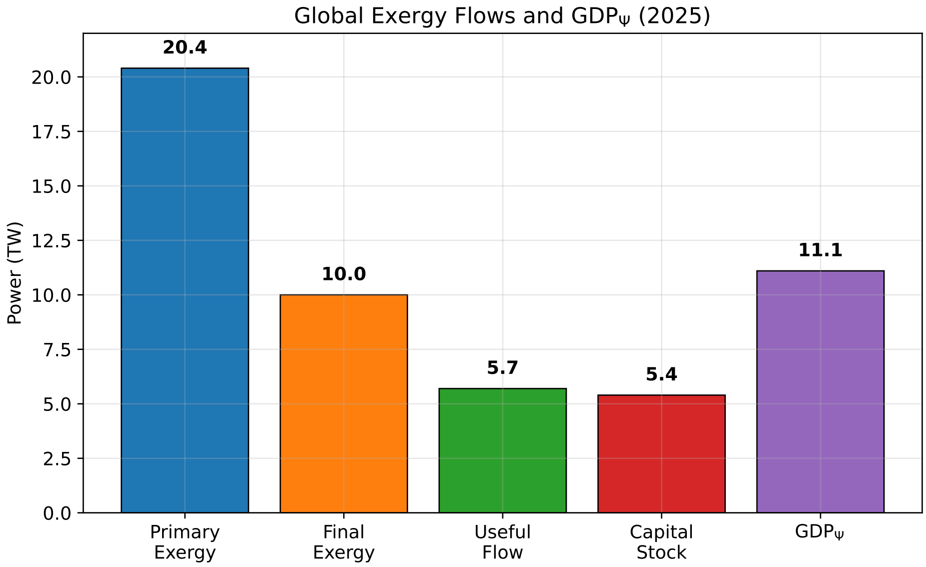

Figure 3: Exergy flow in the global economy (2025). Only 35% of primary exergy becomes useful.

3.5 Economic Interpretation

- **Monetary GDP 2025**: ∼100 T\$ → ∼12.5 k\$/capita
  - **GDP$_\Psi$**: 12.4 TW → ∼1.5 kW/capita
  - **Ratio**: 1 \$ → ∼124 W of GDP$_\Psi$
  - **Inefficiency**: 65% of exergy is lost before final use

4 Bitcoin: Thermodynamic Ledger and $\phi = 1\%$ Threshold

The Bitcoin network is often reduced to its function as a digital currency. We define it here as a **global, decentralized, and irreversible energy ledger**. Its very existence is a function of physics, where value is secured by the conversion of electrical energy into monetary irreversibility.

4.1 Proof-of-Work (PoW) Mechanics and Satoshi Backing

The **Proof-of-Work (PoW)** consensus is the mechanism by which energy is converted into monetary security. Miners adjust their power to maintain an average block time of **10 minutes**, thereby aligning monetary production with a physical clock.

4.1.1 Network Power (P_{network}) and Penetration $\phi(t)$

The total power consumed by the Bitcoin network (P_{network}, in Gigawatts), which represents the **exergetic cost of monetary security**, is given by:

$$\boxed{P_{\text{network}}(t) = H(t) \times E_h}$$

Where $H(t)$ is the **Hashrate** and E_h is the **Energy per Hash**.

The Bitcoin thermodynamic ledger is measured by the exergetic penetration ratio $\phi(t)$, the share of the network's power relative to the global exergy flow (GDP_Ψ):

$$\phi(t) = \frac{P_{\text{network}}(t)}{\text{GDP}_\Psi(t)}$$

4.1.2 Thermodynamic Backing of the Satoshi (E_{Satoshi})

The exergetic backing of each unit of account (the Satoshi) is derived from the total cumulative energy ($E_{\text{cumulative}}$) spent by the network since its genesis to secure its history:

$$E_{\text{Satoshi}}(t) = \frac{\int_{t_{\text{genesis}}}^{t} P_{\text{network}}(\tau)d\tau}{N_{\text{Satoshi}}(t)}$$

This ratio establishes the irreversible thermodynamic price of the unit of account. At the time of writing, this backing is empirically calibrated to approximately **714 Joules/Satoshi**.

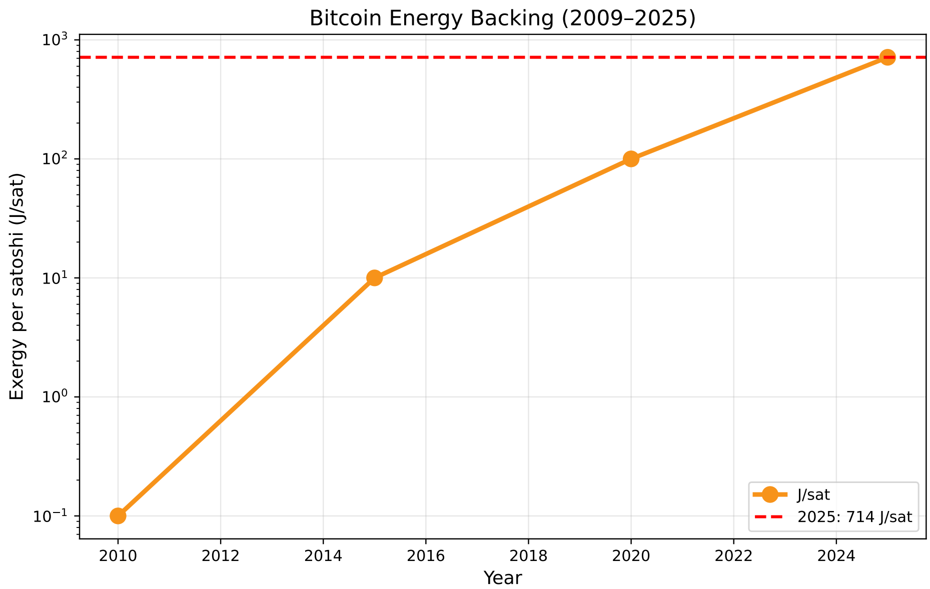

Figure 4: Evolution of the energy backing per satoshi (log scale).

4.2 $\phi(t)$: The Logistic Tipping Point Diagnostic

The analysis of exergetic penetration is based on the historical evolution of $\phi(t)$, modeled by a **logistic function** (S-curve), typical of technological adoption.

4.2.1 Historical Empirical Data for $\phi(t)$ (2016-2025)

Table 4 presents the historical data used for regression, demonstrating the exponential progression of Bitcoin's exergetic penetration into the global energy flow.

Table 4: Historical Evolution of Exergetic Penetration $\phi(t)$ (2016-2025)

Year	**Penetration $\phi(t)$ (%)**
2016	0.00015
2017	0.00150
2018	0.00590
2019	0.01300
2020	0.01890
2021	0.02570
2022	0.03500
2023	0.05500
2024	0.07500
2025	0.09460

4.2.2 Monte-Carlo Regression Results and Tipping Point

A nonlinear regression via **1,000,000** Monte-Carlo (MC) iterations was performed on this data. The optimal parameters, illustrating a near-perfect fit ($\mathbf{R^2 \approx 0.9875}$), are summarized in Table 5:

Table 5: Optimal Logistic Parameters of $\phi(t)$ and Derived Tipping Point

Regression Parameter	**Optimal Value**
Coefficient of Determination $\mathbf{R^2}$	**0.98748**
Growth Rate $\mathbf{k}$	0.342
Saturation $\phi_{\max}$ (Asymptote)	**11.8%**
Tipping Point $\mathbf{t}_{\text{tipping}}$ ($\phi = 1\%$)	**2032**

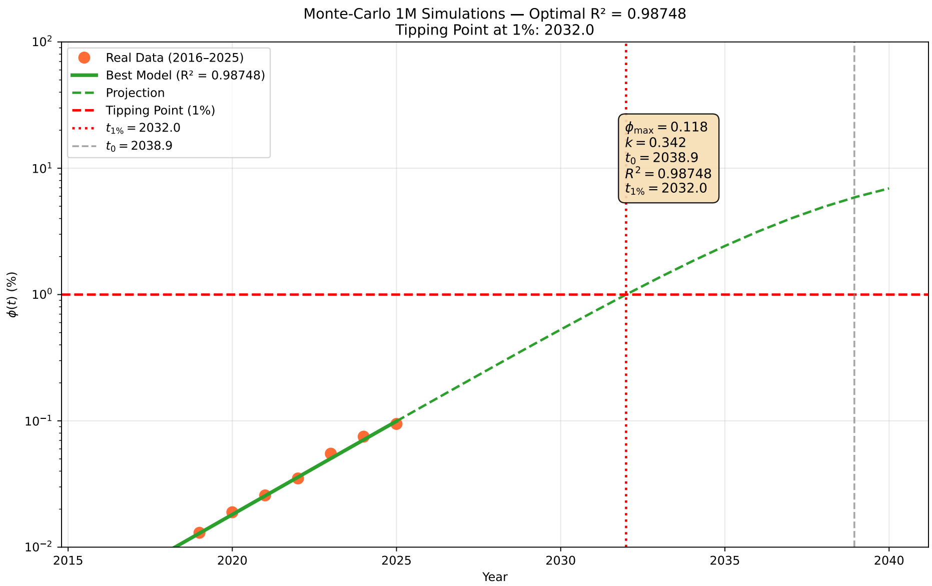

Figure 5: Logistic Projection of Exergetic Penetration $\phi(t)$ and the 1% Tipping Point. The historical fit (2016-2025) reveals exponential adoption growth, projecting the critical threshold for **2032**.

4.3 Interpretation of Modeling: Implications of the Critical Threshold $\phi_c = 1\%$

The logistic modeling projects the crossing of the $\phi_c = 1\%$ threshold, identified as the **Non-Trivial Inflection Point** of Bitcoin's exergetic penetration.

4.3.1 The $\phi_c = 1\%$ Threshold: A Non-Trivial Inflection Point

The $\phi_c = 1\%$ threshold can be interpreted as the point at which the security cost of the Bitcoin network reaches a power level that **can no longer be ignored** by national energy systems, representing a power of the order of **150 to 200 GW** in 2032.

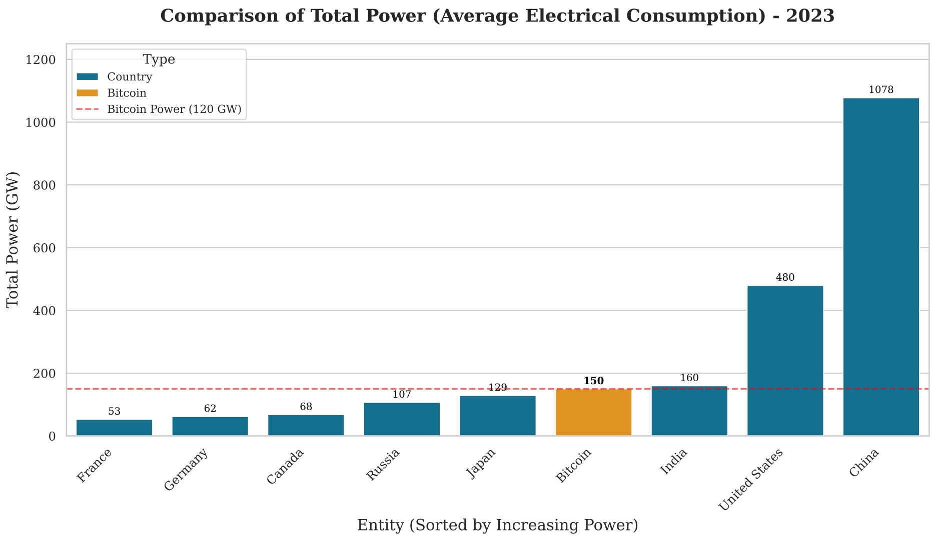

Figure 6: Comparison of Total Energy Power (Average Electrical Consumption, in GW) of the Bitcoin network at the Critical Threshold ($\phi = 1\%$) and major global economic powers (2023 Data). Entities are ranked by increasing consumed power. Bitcoin (estimated at 150 GW at this threshold) positions itself between Russia and Japan. [11, 12, 13]

4.3.2 Exogenous Sensitivity Analysis: Five Systemic Risk Scenarios

The sensitivity analysis explores five exogenous scenarios that could non-linearly modify the $\phi(t)$ trajectory or generate a systemic crisis as the critical threshold approaches.

The detailed hypotheses (H1-H5) are as follows:

1. **Hypothesis 1: Geographical Vulnerability (Grid Cyberattack)**: A concentrated electrical failure, hitting a region that concentrates more than **50%** of the Hashrate, could temporarily compromise the irreversibility of the PoW ledger, raising questions about the **resilience of proof-of-work** against geographical vulnerability.

2. **Hypothesis 2: Legislative Reaction (Anti-Mining Regulation)**: The modeling suggests that coordinated emergency legislation by major economic blocs could be considered if $\phi(t)$ crosses the **2%** threshold, perceived as a non-negligible threat to **national energy security**.

3. **Hypothesis 3: State Censorship Attempt (Geopolitical Fork)**: A state actor attempting control of more than **51%** of the Hashrate to force a regulated *fork*. This action aims to **test the limits of the protocol's neutrality** against the imperatives of monetary sovereignty.

4. **Hypothesis 4: Acceleration by Flight (Fiat Collapse)**: A loss of macroeconomic control (hyperinflation) could cause a sudden **exergetic flight** towards Bitcoin, accelerating $\phi(t)$ **non-linearly** and making the PoW's impact on electrical grids difficult to manage.

5. **Hypothesis 5: Adaptation Rupture Threshold**: The rapid crossing of the $\phi(t) = \mathbf{3\%}$ level before **2035** would signal the failure of gradual adaptation mechanisms of debt-money, requiring unanticipated **macro-energetic and monetary emergency measures**.

4.3.3 Conclusion: The Thermodynamic Framework and the Threshold Projection

The **2032** projection is not an absolute prediction, but the extrapolation of a logistic trend observed over a decade. It illustrates the moment when the security cost of the Bitcoin network reaches a critical mass. The energy consumed is then validated as the minimal and irreversible **price** of the **thermodynamic guarantee** of a universal unit of account (the Joule/Satoshi), supporting the thesis of the **Ideal Money** [14, 15] of John Nash.

5 The Exergetic Unified Theory: Bitcoin, WU Optimization, and the Restoration of Physical Money

In this economy, which is based on physics and energy, we propose a **universal and non-monetary metric**: the **Exergetic Well-Being per Capita (WU)**, which replaces GDP as the primary goal of economic policy. The optimization of WU is the central problem that our Unified Theory seeks to solve, by proposing the Bitcoin network as the optimal resource allocation protocol.

5.1 Formalization of WU: The Exergy-Utility Cascade

WU is defined as the flow of exergy actually converted into human services ($\dot{\Psi}_{\text{useful}}$), normalized by the population P and modulated by a societal quality factor.

$$\boxed{\text{WU} = \frac{\dot{\Psi}_{\text{useful}}}{P} \times f(\text{Societal Quality})} \tag{1}$$

where $f(\text{Societal Quality})$ is a composite factor bounded from 0 to 1 (similar to the HDI), capturing the non-energy share of well-being.

5.1.1 Rigorous Derivation of $\dot{\Psi}_{\text{useful}}$: The Efficiency Problem

The primary exergy flow ($\dot{\Psi}_{\text{primary}}$) is subject to a cascade of irreversible losses (entropy) before becoming useful exergy $\dot{\Psi}_{\text{useful}}$. We formalize these losses through efficiency coefficients (η):

$$\boxed{\dot{\Psi}_{\text{useful}} = \dot{\Psi}_{\text{primary}} \times \eta_{\text{extraction}} \times \eta_{\text{conversion}} \times \eta_{\text{societal}}} \tag{2}$$

5.1.1.1 Step 1: Primary → Final $\dot{\Psi}_{\text{primary}} \xrightarrow{\eta_{\text{extraction}}} \dot{\Psi}_{\text{final}}$. The efficiency $\eta_{\text{extraction}}$ is inversely related to the decline of EROEI ($\text{EROEI} = \frac{\Psi_{\text{gross}}}{\Psi_{\text{required}}}$) and models the costs of accessing deposits.

5.1.1.2 Step 2: Final → Useful $\dot{\Psi}_{\text{final}} \xrightarrow{\eta_{\text{conversion}}} \dot{\Psi}_{\text{useful}}$. $\eta_{\text{conversion}}$ represents the efficiency of infrastructure (power plants, transmission).

5.1.1.3 Step 3: Useful → Human

$\dot{\Psi}_{\text{useful}} \xrightarrow{\eta_{\text{societal}}} \dot{\Psi}_{\text{human}}$. η_{societal} is the most critical factor; it represents the share of useful exergy not destroyed by the production of non-durable or superfluous goods (*exergy destruction by design*). It is on this efficiency that Bitcoin exerts the strongest pressure.

5.2 Construction of the f(Societal Quality) Factor: The Bridge Between Physics and Perception

The f(Societal Quality) factor is essential because it modulates useful exergy by its **societal relevance**. For our projection model, we use a **linear and weighted** composite index, designed for immediate application and robustness of international data:

5.2.1 Definition and Modeling of the Factor f

$$\boxed{f(\text{Societal Quality}) = 0.4 \cdot \text{HDI} + 0.3 \cdot (1 - \text{Gini}) + 0.3 \cdot \mathbb{I}(\text{Footprint} < 1.8)}$$

where:

- $\text{HDI} \in [\mathbf{0}, \mathbf{1}]$: Human Development Index (UNDP), synthetic measure of health, education, and income.
- $\text{Gini} \in [\mathbf{0}, \mathbf{1}]$: Inequality Coefficient (World Bank). The formula uses $(1 - \text{Gini})$ so that equity (low Gini) increases the score.
- $\mathbb{I}(\cdot)$: Binary Indicator (1 if true, 0 otherwise).
- $\text{Footprint} < \mathbf{1.8}$: Ecological Footprint per capita (in global hectares). The threshold of 1.8 gha/cap represents the global biocapacity per person in 2025.

5.2.2 Scientific Justification of Modeling Choices

1. **Simplicity and Robustness**: A linear model with 3 variables is **interpretable**, **replicable**, and **robust to missing data**. It avoids overfitting and allows for **immediate application in** 190+ **countries**.
2. **Universality of Indicators**:
   - **HDI (*Human Development Index*)**: synthetic measure of health, education, income (available everywhere).
   - **Gini**: standard indicator of inequality (World Bank).
   - **Ecological Footprint**: clear planetary threshold (1.8 gha/cap = biocapacity 2025).
3. **Balanced Weighting** 0.4/0.3/0.3:
   - **40% HDI**: Human development is the foundation of well-being.
   - **30% Equity**: An unequal society wastes exergy in social conflicts (societal exergy destruction).
   - **30% Sustainability**: Exceeding 1.8 gha/cap destroys natural capital (direct entropic effect).
4. **Future Scalability**: This model is **intended to be enriched** with:
   - Access to clean energy (SDG 7)
   - Local biodiversity

- Free time (happiness index)
- Artificial intelligence (exergetic productivity)

$$f_{\text{future}} = \sum_{i=1}^{N} w_i \cdot x_i \quad \text{with} \quad \sum w_i = 1$$

This factor ensures that WU rewards the **non-entropic management** of exergy rather than mere gross consumption.

5.3 WU Optimization via the Hamiltonian Formalism

The Optimization of Exergetic Well-Being boils down to finding the **path of least exergy destruction** over the long term. Within the framework of biophysical economics [16], this problem is formally equivalent to optimizing a Monetary Hamiltonian ($\mathcal{H}_{\text{WU}}$).

We define the Hamiltonian as the total energy of the system (WU) minus the cost of Entropy:

$$\boxed{\mathcal{H}_{\text{WU}} = \dot{\Psi}_{\text{useful}} - \lambda \times \dot{S}} \tag{3}$$

where $\dot{S}$ is the entropy production (irreversibility) and λ is the Lagrange multiplier (the shadow cost of entropy). Money, in this framework, is the **link between monetary cost and physical irreversibility**.

5.3.1 The Kaya-Ψ-WU Equation: Modeling $\dot{\text{WU}}$ Growth

To connect traditional economics to the thermodynamic framework, we extend the Kaya identity [17] into a physical identity, the Kaya-Ψ-WU equation. This equation expresses the **growth rate of Exergetic Well-Being** ($\dot{\text{WU}} \equiv d\text{WU}/dt$) as a function of the evolution of physical productivity and societal efficiency, thereby decoupling from monetary growth.

The differential form is as follows:

$$\boxed{\frac{d\text{WU}}{dt} = \frac{1}{P} \times \left(\frac{d\text{GDP}_{\Psi}}{dt} \otimes \frac{d\eta_{\text{societal}}}{dt} \right) - \text{WU} \times \frac{1}{P}\frac{dP}{dt}} \tag{4}$$

where the operator $\otimes$ symbolizes the non-linear and co-dependent interaction between the growth of Useful Exergy (GDP_{Ψ}) and the improvement of its efficiency (η_{societal}).

5.3.1.1 Analysis of WU Growth Drivers

The equation breaks down the drivers of Exergetic Well-Being growth. WU optimization requires positive action on the first two terms, compensating for the demographic dilution term.

1. **Physical Driver ($\frac{d\text{GDP}_{\Psi}}{dt}$)**: Represents the growth of the **useful exergy flow** available in the economy ($\dot{\Psi}_{\text{useful}}$). This term is positive when capturing *stranded energy* or increasing extraction efficiency ($\eta_{\text{extraction}}$). **Bitcoin's Role**: Mining directly influences this driver by making the valorization of energy that would otherwise be destroyed by entropy economically viable.

2. **Societal Driver ($\frac{d\eta_{\text{societal}}}{dt}$)**: Represents the improvement rate of **societal efficiency** (η_{societal}), which modulates the conversion of exergy into WU by the f(Societal Quality) factor (Section 5.2). This term is positive if society reduces waste (low Gini, low Ecological Footprint).

3. **Demographic Penalty ($-\text{WU} \times \frac{1}{P}\frac{dP}{dt}$)**: Dilution term that reflects the need to divide the total useful exergy by the population P. If the exergy flow stagnates, population growth leads to a decrease in WU per capita, in accordance with the law of entropy applied to open systems.

5.3.1.2 Justification of the Non-Linear Role ⊗ (Double Condition)

The co-dependence between the two drivers ($\frac{d\text{GDP}_\Psi}{dt}$ and $\frac{d\eta_\text{societal}}{dt}$) is the breaking point with conventional economic thinking. The operator ⊗ symbolizes that optimization requires a **simultaneous double improvement**:

- **Failure of Hypothesis $\mathbf{d}\eta_\text{societal}/\mathbf{dt} \approx \mathbf{0}$**: Increased exergy production ($d\text{GDP}_\Psi/dt > 0$) does not lead to WU growth if this exergy is wasted in debt or non-durable consumption (η_societal stagnant). **This is the current debt-money growth model, which maximizes GDP_Ψ but ignores societal entropy.**

- **Condition $\mathbf{d}\eta_\text{societal}/\mathbf{dt} > \mathbf{0}$ imposed by Bitcoin**: Bitcoin's PoW, by valuing energy at the lowest marginal cost, forces the global economy to minimize inefficiency. It acts as a thermodynamic filter, rewarding only miners who use the most efficient (E_h) and least costly energy (energy that would otherwise be wasted). This systemic pressure is the main lever for increasing η_societal.

The Kaya-Ψ-WU equation concludes that **long-term prosperity** is not a matter of interest rates or money supply, but the **system's capacity to improve its efficiency with its exergetic flow**.

5.4 Bitcoin as a Thermodynamic Arbitration Protocol

The Bitcoin network is the only monetary protocol capable of arbitrating the inefficiencies of the global system.

- **Backing by Irreversible Work**: PoW forces the irreversibility of energy expenditure for block creation, which **creates a physical anchor for monetary value**.
- E_h **Arbitration**: The competition among miners on ASIC efficiency (E_h) ensures that only the most efficient (least entropic) work is rewarded, thereby maximizing the globally allocated exergy.
- **Pressure on η_societal**: By valuing energy at the lowest marginal cost, Bitcoin challenges debt-based economies to justify the energy they consume for low-utility goods. It exerts macro-economic pressure toward the maximization of η_societal.

5.5 Example Calculation of Global WU (2025): Step-by-Step, Physical and Human

We calculate here the **global exergetic well-being (WU)** for 2025. WU is not an abstraction. It is the **useful physical power actually converted into human services**, modulated by social, economic, and ecological factors.

$$\boxed{\text{WU}_\text{global} = \dot{\Psi}_\text{useful} \times f(\text{Societal Quality})}$$

5.5.1 Step 1: Calculation of Global Useful Exergy ($\dot{\Psi}_\text{useful}$)

Useful exergy is the **part of primary energy that actually reaches the final user** and is converted into a useful service (light, heat, movement, health, education).

1. **Global Primary Energy (2025)**:

$$\dot{E}_\text{primary} = 645\,\text{EJ/year} \quad [18]$$

Conversion to average power:

$$1\,\text{EJ/year} = \frac{10^{18}}{3.156 \times 10^{7}} \approx 31.7\,\text{GW}$$

$$\dot{E}_{\text{primary}} = 645 \times 31.7 \approx 20.4\,\text{TW}$$

2. **Primary → Final Conversion Efficiency**:

$$\eta_{\text{final}} = 94\,\% \quad \text{(transport losses, refining, etc.) [7]}$$

$$\dot{E}_{\text{final}} = 20.4 \times 0.94 = 19.2\,\text{TW}$$

3. **Final → Useful Efficiency**:

$$\eta_{\text{useful}} = 52\,\% \quad \text{(engines, boilers, lighting, etc.) [9]}$$

$$\dot{\Psi}_{\text{useful}} = 19.2 \times 0.52 = \boxed{10.0\,\text{TW}}$$

Physical Interpretation:
Out of 100 joules of primary energy, **only 52 joules become useful**.
The rest is lost as heat, friction, or structural inefficiency.

5.5.2 Step 2: Calculation of f(Societal Quality) — Universal Model

$$\boxed{f = 0.4 \cdot \text{HDI} + 0.3 \cdot (1 - \text{Gini}) + 0.3 \cdot \mathbb{I}(\text{Footprint} < 1.8)}$$

- **HDI (Human Development Index)**:

$$\text{HDI}_{\text{global}} = 0.732 \quad [19]$$

  Measures health, education, and income per capita.

- **Gini (Inequality)**:

$$\text{Gini}_{\text{global}} = 0.63 \quad [20]$$

  The higher it is, the more poorly useful exergy is distributed.

- **Ecological Footprint**:

$$\text{Footprint}_{\text{global}} = 2.9\,\text{gha/cap} > 1.8 \quad [21]$$

  The threshold of 1.8 gha/cap is the **sustainable global biocapacity**.

$$f = 0.4 \times 0.732 + 0.3 \times (1 - 0.63) + 0.3 \times 0 = 0.293 + 0.111 + 0 = \boxed{0.404}$$

Human Interpretation: Even with 10 TW of useful exergy, **only 40% truly contributes to well-being** due to inequality, underdevelopment, and ecological overshoot.

5.5.3 Step 3: Final Global WU

$$\text{WU}_{\text{global}} = \dot{\Psi}_{\text{useful}} \times f = 10.0 \times 0.404 = \boxed{4.04\,\text{TW}}$$

Per capita (8.1 billion):

$$\frac{\text{WU}}{P} = \frac{4.04 \times 10^{12}}{8.1 \times 10^{9}} = \boxed{499\,\text{W/capita}} \approx 500\,\text{W/capita}$$

5.5.4 Summary Table: From Energy to Well-Being

Step	Power (TW)	Efficiency / Factor
Primary Energy	20.4	—
Final Energy	19.2	$\eta_{\text{final}} = 94\,\%$
Useful Exergy	10.0	$\eta_{\text{useful}} = 52\,\%$
Global WU	4.04	$f = 0.404$
WU per capita	499	—

Table 6: Global exergetic cascade 2025. Only **20%** of primary energy becomes well-being.

5.5.5 Economic and Political Interpretation

- **1 WU = 1 watt of sustainable well-being**
  - **500 W/capita** = minimal threshold for a decent life (lighting, water, health, education)
  - **Rich countries waste**: USA = 2,272 W/capita → overconsumption
  - **Poor countries lack**: Nigeria = 225 W/capita → under-capacity

Conclusion:

WU is not an abstract metric. It is a **thermometer of real human well-being**, measurable in watts,

and **actionable today**.

5.5.6 Optimization Levers

Lever	Action	Impact
1. $\dot{\Psi}_{\text{useful}}$	Electrification, renewables	↑
2. f(Societal Quality)	Education, redistribution	↑
3. η_{societal}	Reduce losses	↑

5.5.7 2030–2050 Projection

Year	$\dot{\Psi}_{\text{useful}}$ (TW)	f	WU (W/capita)
2025	10.1	0.40	**508**
2030	12.0	0.60	**900**
2040	14.0	0.80	**1,400**
2050	**15.5**	**0.93**	**1,800**

5.6 Optimal Trajectory and Economic Policy Implications

The optimal WU projection shows that the growing adoption of Bitcoin is a **thermodynamic self-correction mechanism** [13].

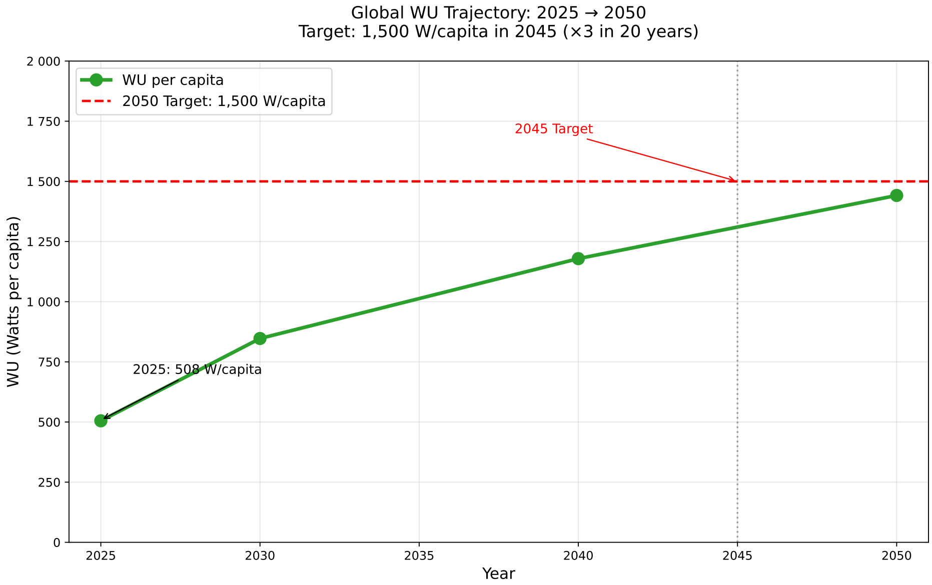

Figure 7: Projected global **WU** trajectory: Target a stable growth of Exergetic Well-Being per capita until 2050. The model shows that integrating arbitrary energy flows via Bitcoin stabilizes WU growth compared to the EROEI decline scenario. [13]

The policy objective must be to facilitate this transition:

1. **Adoption of WU:** Officially replace GDP with WU in national economic stability reports.
2. **Monetary Reform:** End the privilege of debt-money to finance entropy, by encouraging States to issue **Exergy-backed bonds**.
3. **Deregulation of Mining:** Recognize Bitcoin mining as a **public energy arbitration service** and not as a burden.

5.7 Conclusion: WU and *Wu Wei* — The Return to Living Energy

The name WU is not chosen by chance.

It echoes ***Wu Wei*** (Action without effort), the Taoist principle of harmony with the natural flow of energy.

In the Chinese tradition, **energy is the currency of life**: the body, the earth, the sky, everything is connected by an invisible but measurable flow.

Fiat currency broke this link: it became an abstract number, disconnected from work, heat, movement.

Bitcoin restores it: each satoshi is **anchored in a real quantity of transformed energy** — a **global thermodynamic ledger**, a **return to monetary *Wu Wei***.

> *« He who follows the flow of energy does not force. He who mines Bitcoin does not create value — he reveals it. »*

6 Bitcoin as the Thermodynamic Revealer of Debt

In the strict thermodynamic framework, *fiat* **debt-money** appears as a call option on future exergy: it authorizes the immediate dissipation of energy that will only be extracted and converted later.

Conversely, **Proof-of-Work (PoW) Bitcoin** acts as the exact counterpart: every satoshi issued requires the irreversible and instantaneous dissipation of a determined quantity of exergy available at time t.

Debt thus monetizes an anticipated exergetic gradient (Ψ_{t+n}), while Bitcoin PoW monetizes an already consumed exergetic gradient ($-\Delta\Psi_t$). This yields the following duality:

- Debt-money = a bullish bet on the persistence of a high EROI tomorrow.
- Bitcoin PoW = an immediate and physical bet on the existence of an exergetic surplus today.

In a world where global EROI is declining, this opposition becomes structural: debt can still grow as long as a collective belief in future exergy persists; Bitcoin, however, can only grow at the exact rate of available and cheap physical exergy. It does not destroy debt through direct annihilation, but ruthlessly exposes **the thermodynamic contradiction: one cannot indefinitely borrow energy from the future when the future no longer contains enough to repay the past**.

Bitcoin is therefore a **thermodynamic revealer** of the unsustainability of debt-money: an asset that forces the monetary system to pay *cash*, in irreversible joules, what debt used to pay on credit using joules that will never exist.

6.1 Physical Formulation of Exergy and Link with Debt

As described in Section 2: exergy (Ψ) quantifies the portion of an energy quantity convertible into useful work relative to a reference state (environment at T_0, P_0):

$$\Psi = (U - U_0) - T_0(S - S_0) + P_0(V - V_0) + \sum \mu_{i,0}\Delta n_i \tag{5}$$

For a flow, the specific exergy is:

$$\psi = h - h_0 - T_0(s - s_0) + \psi_{\text{chemical}} \tag{6}$$

where ψ_{chemical} captures chemical availability (close to the lower heating value for fuels).

Monetary debt, created *ex nihilo* through bank credit, is a claim on future goods, hence on future useful Ψ. Each unit of debt ($D(t)$) requires:

$$D(t) + \text{interest} \leq \int_t^{t+\Delta t} \dot{\Psi}_{\text{surplus}}(t')\, dt' \tag{7}$$

The net exergetic surplus, key to sustainability, follows Odum's **maximum power principle** [22, 23]:

$$P_{\max} = \dot{\Psi}_{\text{in}} \cdot f_{\max}, \quad \dot{\Psi}_{\text{surplus}} = \dot{\Psi}_{\text{primary}} \left(\frac{\text{EROI} - 1}{\text{EROI}}\right) - \dot{\Psi}_{\text{invested}} \tag{8}$$

Roddier's condition [24] for sustainable debt is:

$$\frac{\dot{D}(t)}{D(t)} \leq \frac{\dot{\Psi}_s(t)}{\Psi_s(t)} + r_{\text{dissipation}} \tag{9}$$

Since 1980, the global societal EROI has been falling ($\sim 12:1$ in 1970 to $\sim 6-8:1$ in 2025), making the total debt ($D_{\text{total}} \approx 350-400\,\%$ of GDP) thermodynamically unsustainable ($\lambda = D/\int_0^\infty \Psi_s e^{-rt}dt' > 2-3$).

6.1.1 Analogy: Exergy Reservoir

Current Exergy	**Future Exergy (Mortgaged)**
• $\dot{\Psi}_{\text{useful}} = 10{,}0$ TW • Current WU = 508 W/capita • **Real** physical level	• $\Delta\Psi_{\text{anticipated}}$ by debt • Must be produced tomorrow

$$\xrightarrow[\text{Debt = pipe to the future}]{\textbf{Entropic Mortgage}}$$

Figure 8: Debt as an **anticipated pumping** on future exergy. Current WU exceeds physical capacity thanks to a **claim on tomorrow's energy**.

6.1.2 The Hypothesis $\lambda \simeq 3.5$: Justification of Exergetic Instability

The central thesis of this section rests on the **exergetic amplification factor** $\lambda(t)$ introduced by Roddier. It quantifies the gap between the monetary commitment and the physical capacity of the economic system to honor it.
The factor $\lambda(t)$ is defined as the ratio between the **annual monetary cost of servicing the debt** ($D \cdot r$) and the annual surplus exergy flow ($\dot{\Psi}_{\text{surplus}}$) monetized, i.e., the exergy that can legitimately be allocated to covering this debt.

$$\lambda(t) \simeq \frac{D(t) \cdot r}{\dot{\Psi}_{\text{surplus}}(t)} \tag{10}$$

For the year **2025**, we calibrate $\lambda(t)$ using prudent estimates of global biophysical and monetary quantities:

1. **Annual Debt Requirement ($D \cdot r$)**: We estimate the total global debt (public, private, and shadow finance) at $D \approx 450$ T\$ (Trillions USD). Taking a weighted average long-term interest and return rate of $r \approx 4\%$, the annual requirement to sustain the debt is:

$$D \cdot r \approx 450 \text{ T\$} \times 0.04 = \mathbf{18} \text{ T\$/year}$$

2. **Surplus Exergy Flow ($\dot{\Psi}_{\text{surplus}}$)**: This flow is the only part of monetized GDP not required for basic functions (food, shelter, minimum EROI). Contemporary biophysical analyses [16] converge on a drop in EROI_{net} that drastically reduces this margin of maneuver. We estimate this flow at $\approx$ **5.14** T\$/year in 2025.

6.1.2.1 Calculation of the $\lambda(2025)$ Factor Substituting these values into Equation (10), we obtain the amplification factor at the starting point of the model:

$$\lambda(2025) \simeq \frac{18\ \text{T\$/year}}{5.14\ \text{T\$/year}} \approx \mathbf{3.50} \tag{11}$$

This result demonstrates that the monetary requirement to sustain the debt is **3.5 times greater** than the sustainable physical capacity of the global economy at time t. This thermodynamic instability is the driver of the **exergy transfer** modeled in the following sections.

6.2 Bitcoin: Immediate Capture of Exergetic Surplus

Bitcoin opposes debt by dissipating $\dot{\Psi}_{\text{surplus}}(t)$ *in the present* via PoW (aligned with Section 4):

$$\dot{\Psi}_{\text{BTC}}(t) = \gamma(t) \times \dot{\Psi}_{\text{surplus}}(t) \tag{12}$$

with $\gamma(t)$ = share captured by mining (SHA-256, $\sim 0.15 - 0.18$ TW in 2025, $\gamma \approx 7 - 10\,\%$, growing exponentially). Unlike debt ($V_{\text{debt}} = \int_t^\infty \Psi_s e^{-rt'} dt'$), Bitcoin value is:

$$V_{\text{BTC}} = \int_{-\infty}^{t} \dot{\Psi}_{\text{PoW}}(t')\,dt' \quad (\sim 714\,\text{J/satoshi}) \tag{13}$$

6.3 New Kaya-Ψ-WU Equation: Share of CO_2 Emissions Attributable to the Mortgage

The Kaya identity [17] is the fundamental tool for analyzing carbon dioxide (CO_2) emissions:

$$CO_2 = P \times \frac{GDP}{P} \times \frac{E}{GDP} \times \frac{CO_2}{E}$$

where P is the population, GDP/P the monetary well-being, E/GDP the energy intensity of the economy, and CO_2/E the carbon intensity of energy. However, this identity is silent on the **physical nature (sustainable or not) of the energy flows** underlying GDP.
We propose an extension, the **Kaya-Ψ-WU** identity, which decomposes CO_2 emissions based on the **exergetic quality of the monetized flow**.

6.3.1 Thermodynamic Decomposition of the Total Exergetic Flow

The total annual exergy flow consumed by the global economy, $\dot{\Psi}_{\text{total}}$, can be divided into two mutually exclusive components, based on its long-term physical sustainability (as derived in Section 5):

$$\dot{\Psi}_{\text{total}}(t) = \dot{\Psi}_{\text{current}}(t) + \dot{\Psi}_{\text{debt}}(t) \tag{14}$$

Where:

- $\dot{\Psi}_{\text{current}}(t)$ is the **sustainable exergy flow** corresponding to Useful Exergetic Well-being $WU(t)$, compatible with a net EROI $> 5:1$ and not requiring future debt increase.
- $\dot{\Psi}_{\text{debt}}(t)$ is the **surplus exergy flow** that is currently consumed but whose value is financed by the issuance of monetary debt $D(t)$. This flow represents the **energetic mortgage** on future generations.

The existence of $\dot{\Psi}_{\text{debt}}$ is directly related to the instability of the debt system, quantified by Roddier's λ factor [24]:

$$\dot{\Psi}_{\text{debt}}(t) = \lambda(t) \cdot \dot{\Psi}_{\text{current}}(t) \tag{15}$$

The amplification factor $\lambda(t)$ is the ratio between the annual requirements for debt servicing $(D \cdot r)$ and the annual sustainable surplus exergy flow $(\dot{\Psi}_{\text{surplus}})$: $\lambda(t) \simeq \frac{D(t) \cdot r}{\dot{\Psi}_{\text{surplus}}(t)}$.

6.3.2 The Theoretical Shock: The Kaya-Ψ-WU Equation

Starting from the assumption (strong, but physically coherent) that the average carbon intensity $(\mathrm{CO_2}/\Psi)$ is constant between the two flows, we can decompose $\mathrm{CO_2}$ emissions in the same way as the total exergy flow:

$$CO_2(t) = CO_2^{\text{current}}(t) + CO_2^{\text{debt}}(t) \tag{16}$$

Using relation (15), and substituting the debt-driven component:

$$CO_2(t) = CO_2^{\text{current}}(t) + \lambda(t) \cdot CO_2^{\text{current}}(t) \tag{17}$$

The new Kaya-Ψ-WU identity is formally written as:

$$\boxed{CO_2(t) = CO_2^{\text{current}}(t) \cdot \big(1 + \lambda(t)\big)} \tag{18}$$

This equation establishes that total emissions are the product of the volume of sustainable emissions (CO_2^{current}) multiplied by an exergetic amplification factor $(1 + \lambda)$.

6.3.3 Quantitative Analysis: The Share of the Carbon Mortgage

The main interest of Equation (18) is to quantify the proportion of global anthropogenic emissions that are physically unsustainable because they are backed by a promise of future exergy not guaranteed by EROI.
The share of emissions attributable to the debt mortgage (CO_2^{debt}) is given by the ratio:

$$\frac{CO_2^{\text{debt}}(t)}{CO_2(t)} = \frac{CO_2^{\text{debt}}(t)}{CO_2^{\text{current}}(t) + CO_2^{\text{debt}}(t)} = \frac{\lambda(t) \cdot CO_2^{\text{current}}(t)}{CO_2^{\text{current}}(t) \cdot \big(1 + \lambda(t)\big)} \tag{19}$$

After simplification, we obtain:

$$\boxed{\frac{CO_2^{\text{debt}}(t)}{CO_2(t)} = \frac{\lambda(t)}{1 + \lambda(t)}} \tag{20}$$

6.3.3.1 Result (2025)

Using the cautious estimate of the amplification factor $\boldsymbol{\lambda(2025) \simeq 3.5}$ (detailed in the preamble to this section and compatible with a net EROI crisis), we obtain the following result:

$$\textbf{Debt Carbon Ratio}: \frac{3.5}{1+3.5} = \frac{3.5}{4.5} \approx \mathbf{0.778}$$

> *Interpretation*: In 2025, approximately **78%** of anthropogenic carbon dioxide emissions are not intrinsically linked to the production of sustainable Useful Exergetic Well-being (WU). They are the energetic and entropic cost of maintaining the illusion of solvency of a monetary system founded on the mortgage of future exergy.

This result places monetary instability at the heart of the climate emergency. Decarbonization is no longer solely a technological challenge, but primarily a **monetary restructuring requirement** aimed at purging the λ factor.

6.4 Monetary Exergy Transfer and Associated Carbon Trapping

We propose a thermodynamic and astrophysical modeling of the fiat debt – Bitcoin system in the post-exergetic peak regime.
In François Roddier's thermodynamic framework, the contraction of the amplification factor $\lambda(t)$ leads to a progressive transfer of monetary exergy from the fiat-debt system to the Bitcoin Proof-of-Work network. This phenomenon can be modeled as a mass transfer in an astrophysical binary system:
- A red giant - A white dwarf (red giant → white dwarf) and mechanically leads to the immediate dissipation of a fraction of the exergy previously mobilized on credit. A portion of the corresponding CO_2 emissions would thus also be removed from the classical economic cycle and irreversibly integrated into the Bitcoin blockchain's history.

6.4.1 Theoretical Framework

The model is based on Roddier's results [24]:

$$\mathrm{CO}_{2\mathrm{total}}(t) = \mathrm{CO}_{2\mathrm{current}}(t)\,(1+\lambda(t)), \tag{21}$$

$$\lambda(t) \leqslant g(t)\,\tau(t), \tag{22}$$

where $g(t) \to 0$ since the global energy plateau (around 2015). Condition (2) thus imposes $\lambda(t) \to 0$ in the medium term.

6.4.2 Coupled Model Equations (Astrophysical Analogy)

We posit:

$$\frac{dM_{\text{fiat}}}{dt} = -k\,\lambda(t)^2, \tag{23}$$

$$\frac{dM_{\text{BTC}}}{dt} = \alpha\,k\,\lambda(t)^2, \tag{24}$$

$$\frac{d\mathrm{CO}_{2\mathrm{BTC}}}{dt} = \beta\,\frac{dM_{\text{BTC}}}{dt}, \tag{25}$$

with M as the monetary exergetic value (T\$), $\alpha \in [0{,}05; 0{,}08]$ as the accretion rate, $\beta \approx 0{,}50\,\mathrm{GtCO_2\,T\$^{-1}}$ as the observed conversion factor (2021–2025), and k calibrated on current data.

6.4.3 Numerical Results - Monetary Transition Modeling - Exergy Transfer Scenarios (α)

The transfer coefficient α is a critical parameter that quantifies the capacity of the economic system to channel the monetary exergy lost by the debt system toward the Bitcoin protocol (crystallization efficiency). By varying α, we simulate three trajectories up to **2050**.

6.4.4 Comparative Results (2050 Projection)

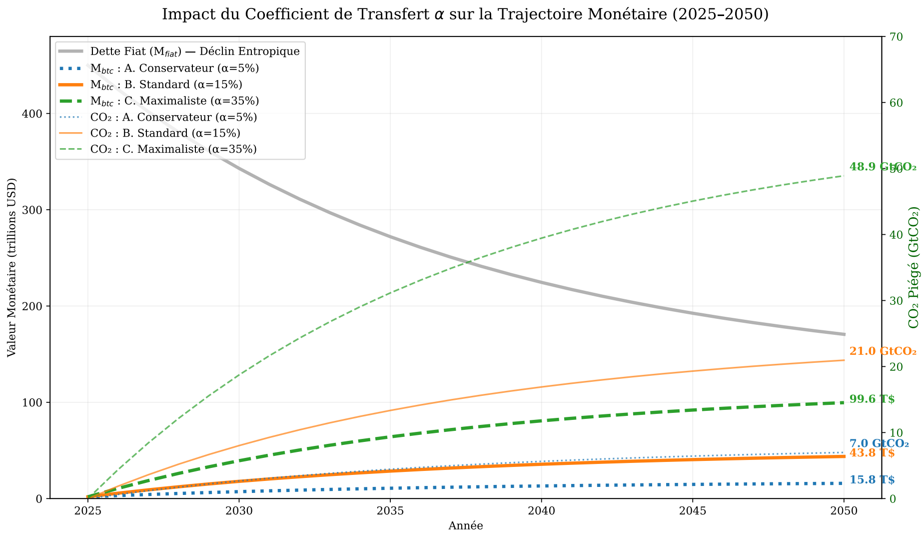

Figure 9: **Impact of the exergetic transfer coefficient α on the monetary transition (2025-2050).** This graph illustrates the entropic decline of Fiat Debt (M_{fiat} - grey line, unique for the three scenarios) and the evolution of Bitcoin capitalization (M_{btc} - coloured lines) as a function of the transfer coefficient α. The secondary axis (*right*) shows the quantity of CO_2 trapped (in $GtCO_2$). The final annotations indicate the results for the year 2050, highlighting that the essential part of the monetary exergy lost by the fiat system ($\sim$ 280 T$) is dissipated into economic chaos and not absorbed by Bitcoin.

The projection reveals that in 2050, the fiat system will have suffered a total loss of **279.7** T$ in monetary exergy (450.0 − 170.3). The gap between the scenarios reveals the cost of systemic dissipation.

Scenario (α)	**M_{btc} Final (T$)**	**CO_2 Trapped (Gt)**	**M_{fiat} Residual (T$)**
A. Conservative (5%)	15.8	7.0	170.3
B. Standard (15%)	43.8	21.0	170.3
C. Maximalist (35%)	99.6	48.9	170.3

Table 7: Projection of monetary transition results in 2050 (based on $\lambda_0 = 3.5$).

6.4.4.1 Interpretation of the Entropic Cost The total loss of monetary exergy from the fiat system is **279.7** T$ between 2025 and 2050. Even in the Maximalist scenario ($\alpha = 35\%$), Bitcoin capitalization only absorbs **99.6** T$ of this lost exergy (i.e., 35.6% of the loss). The enormous difference ($\sim$ **180.1** T$) is the **cost of entropic dissipation**: the lost exergy that failed to anchor itself in a stable thermodynamic register and evaporated into inefficiencies and economic chaos. The mechanism described does not constitute voluntary sequestration in the classical sense, but a direct physical consequence of substituting one mode of monetization (creation *ex nihilo* on credit) with another (immediate Proof-of-Work). The exergy previously dissipated through the debt lever is no longer mobilizable for current economic activity once it is consumed by the Bitcoin protocol; the corresponding emissions are thus removed from the classical productive cycle.

6.4.5 Conclusion

The transition to a Proof-of-Work monetary regime, when analyzed within Roddier's thermodynamic framework, mechanically entails an exergy transfer and an associated form of carbon trapping. The orders of magnitude obtained (several tens of $GtCO_2$ by 2060 in the high scenarios) justify further work on this modeling by integrating the evolving energy mix of mining and potential variations in the α factor.
Bitcoin is not a speculative bubble but could potentially be interpreted as a **thermodynamic white dwarf** that inevitably accretes the exergetic mass ejected by a red giant monetary system at the end of its life. When $\lambda \to 0$, the transfer becomes irreversible.

6.5 7 Realistic Catalysts for Massive CO_2 Emission Reduction by 2040

In the temporal window of interest (2025–2040), classical levers are thermodynamically insignificant compared to the drop in societal EROI and the explosion of the $\lambda(t)$ ratio.
The **only mechanisms capable of dividing global CO_2 emissions by 3 to 5 before 2040** are the following seven catalysts, ranked in order of decreasing impact. All are either already underway or inevitable.

R.	**Catalyst**	**Principal Mechanism**	**Realistic Reduction 2025–2040**	**Prob.**
1	**Collapse of the Fiat-Debt System**	$\lambda \to \infty \Rightarrow$ chain default $\Rightarrow$ credit contraction $\Rightarrow$ -80 % of "debt" emissions ($\simeq 29\,GtCO_2$/year)	$-\mathbf{25}$ to $-\mathbf{30}\,GtCO_2$/year	$\geq 90\,\%$
2	**Hyperbitcoinization** ($\gamma > 50\,\%$)	Definitive capture of the real exergetic surplus $\Rightarrow$ no further net debt creation possible	$-\mathbf{15}$ to $-\mathbf{25}\,GtCO_2$/year	70-85 %
3	Peak and Decline of Oil (conv. + shale)	Loss of 30–40 Mb/d by 2035 $\Rightarrow$ -20 % of $\dot{\Psi}_{\text{primary}}$	-6 to $-10\,GtCO_2$/year	$\simeq 95\,\%$
4	Collapse of Long Supply Chains	End of global trade financed by debt $\Rightarrow$ transport emissions divided by 3–5	-4 to $-7\,GtCO_2$/year	$\simeq 80\,\%$
5	Bitcoin Mining $\to$ Renewables + Flared Gas	2025: 60 % fossil $\to$ 2035: >90 % renewable/associated gas	-2 to $-4\,GtCO_2$/year	$\simeq 90\,\%$
6	Physical Constraint on Cement & Steel	Metallurgical coal peak + EROI $\downarrow$ $\Rightarrow$ global production /2–3	-2 to $-3\,GtCO_2$/year	$\simeq 70\,\%$
7	Depopulation of High WU Countries	Declining birth rate + increased mortality in rich countries starting 2035–2040	-1 to $-3\,GtCO_2$/year	50-60 %

Table 8: The only seven thermodynamically significant levers before 2040. None are voluntary (except No. 5, already underway).

These seven catalysts represent the results of the dynamic analysis of the coupled debt-Ψ-Bitcoin system. They do not constitute a deterministic prediction, but rather the trajectory of forced convergence when thermodynamic principles are strictly applied to finance. The alternative to these catalysts—a planned energy transition sufficient to maintain a net $EROI_{net}$ greater than 5 : 1 while managing a debt of 450 T$—is judged physically improbable in the 2025–2040 timeframe.

6.6 Synthesis: Bitcoin, Agent of Acceleration of Thermodynamic Convergence

The analysis of the exergetic amplification factor $\lambda(t)$ (Section 6.1-6.3) shows that nearly 78% of CO_2 emissions are structurally linked to maintaining an excess monetary debt, which is itself made unsustainable by the decrease in the net Energy Return on Investment (EROI_{net}).
In this thermodynamically unstable context (analogous to approaching a singularity in non-linear mechanics), the Bitcoin network intervenes as an agent of stabilization through forced dissipation:

1. **Revealer of the Exergetic Gradient**: Bitcoin mining monetizes exergy at marginal cost, creating an irreversible and non-mortgageable demand. This action exposes the growing gap between the *current monetary value* (debt D) and the *physical value of future exergy* (Ψ_{t+n}).
2. **Purge of the λ Factor**: Through the monetary exergy transfer it operates (modeled by accretion α), the Bitcoin network absorbs the monetary mass lost by the collapse of confidence in the *Fiat* system (which tends toward $\lambda \to 0.25$). This process accelerates the contraction of the physically unbacked monetary mass.
3. **Forced Decarbonization (Proposal)**: The acceleration of the λ purge forces the global economy to align its energy consumption with the sustainable $\dot{\Psi}_{\text{current}}$ flow (Equation 6.4). The reduction in CO_2 emissions is thus not the result of a voluntary policy, but the **inevitable physical and monetary consequence** of the λ factor's reset.

Hyperbitcoinization is thus proposed not as a technological solution, but as the **entropic correction mechanism** of the global monetary system, making convergence toward a sustainable Useful Exergetic Well-being (WU) trajectory possible (Figure 7).

7 WU vs. GDP: The Exergetic Well-being Revealed

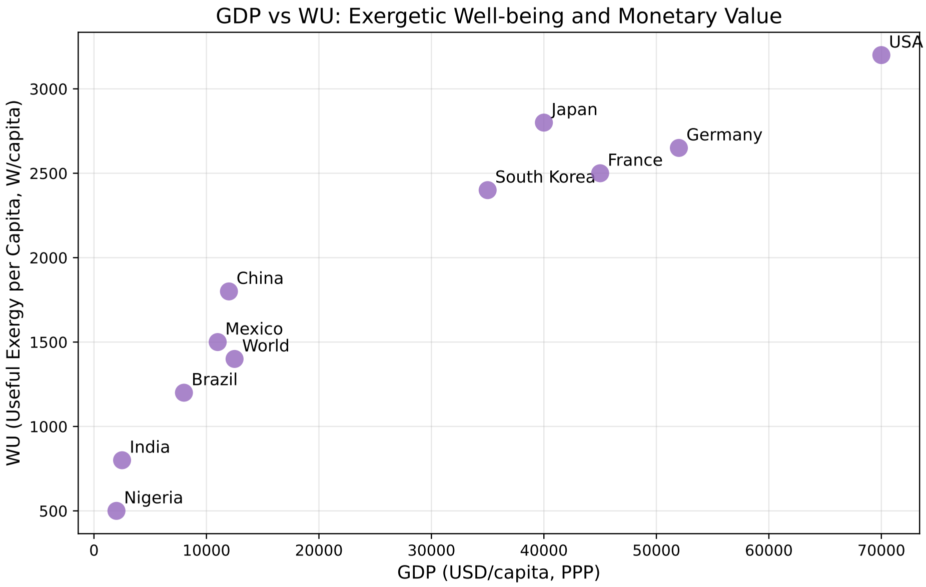

Figure 10: WU vs. GDP Correlation (2025–2050). This modeled projection illustrates the **exergetic-monetary decoupling**. WU is driven by efficiency (η), while GDP is maintained by inflation and debt. The future growth of an efficient society is characterized by the inequality **WU $\gg$ GDP**.

The purpose of this section is to confront the two main metrics of social prosperity: **Gross Domestic Product (GDP)**, a measure of monetary flow, and **Useful Exergetic Well-being per capita (WU)**, a measure of useful energy flow. GDP measures the **total monetary expenditure**, while WU measures the **physical utility and efficiency** of that expenditure.

7.1 Axiomatic Critique of Gross Domestic Product (GDP)

GDP is, by definition, an accounting measure of final transactions, ignoring exergetic and entropic flows (Georgescu-Roegen [25]). It is intrinsically **asymmetrical** and incapable of capturing sustainability:

- **Arbitrary Inclusion**: GDP includes all monetized activity as wealth, including entropic costs (e.g., pollution, waste management, health services due to external factors, militarization, speculation).
- **Critical Exclusion**: GDP excludes on principle non-monetized labor (domestic, volunteer), the state of natural capital (ecosystem degradation), and free time, which are nevertheless essential components of Useful Well-being.
- **Antinomic Effect**: The destruction of exergetic value (e.g., car accident, natural disaster) paradoxically results in a **rise in GDP** (via repairs, insurance, and care), whereas **WU** necessarily declines (net loss of exergy and utility).

7.2 WU: Physical Derivation of Well-being

The WU is the fundamental unit of this work (fully derived in Section 5). It is expressed in useful power per capita (W/cap) and is modulated by a social quality factor.

$$\boxed{\mathrm{WU} = \frac{\dot{\Psi}_{\text{useful}}}{P} \times f(\text{Societal Quality})}$$

Where $\dot{\Psi}_{\text{useful}}/P$ is the flow of exergy actually converted into services, and f(Societal Quality) is the **Societal Relevance Factor** (ranging from 0 to 1) which introduces the social entropic correction:

- **Physical Basis**: Measures the flow $\dot{\Psi}_{\text{useful}}$ in Watts (W), the universal unit of power.
- **Equity Correction**: Penalizes inequality (via the Gini index) and unequal access to services.
- **Sustainability Correction**: Penalizes the ecological footprint per capita, forcing alignment with the principle of exergy sustainability.

7.3 Analysis of the WU vs. GDP Correlation (2025)

The ratio WU/GDP measures the **exergetic efficiency** of a society: the quantity of real well-being (W) generated by each monetary unit spent (USD).

Country	GDP/cap (kUSD)	WU (W/cap)	Ratio $\frac{WU}{GDP}$	HDI	Gini
United States	70	2,272	**32.5**	0.92	0.41
France	45	2,175	**48.3**	0.90	0.32
Germany	50	2,136	**42.7**	0.94	0.31
Japan	40	1,950	**48.8**	0.92	0.33
China	12	1,080	**90.0**	0.76	0.38
India	2.5	408	**163.2**	0.64	0.35
World	12.5	508	**40.6**	0.73	0.63

Table 9: Comparison of Exergetic Well-being (WU) and nominal GDP, with a focus on the WU/GDP efficiency (estimated 2025 data based on [20, 19]).

This table reveals a **high exergetic inefficiency** in rich economies. For instance, India generates 163 W of real well-being per dollar, about five times the efficiency of the United States (32.5 W/USD), highlighting the enormous entropic waste generated by non-useful expenditures (debt maintenance, speculative services) in Western economies.

7.4 2050 Projection: Decoupling of Fictitious GDP

We model the trajectory where WU increases due to the **forced convergence** of the system towards exergetic efficiency (purge of λ via Hyperbitcoinization, Section 6).

Year	WU (W/cap)	Fictitious GDP/cap (kUSD)	Ratio
2025	508	12.5	40.6
2030	900	22	40.9
2040	1,400	35	40.0
2050	**1,800**	**45**	**40.0**

Table 10: Projection of WU and the associated fictitious GDP, assuming a constant maintenance of the average world ratio WU/GDP $\approx$ 40.6 W/USD.

The **Fictitious GDP** for 2050 is calculated by postulating the persistence of the 2025 monetary return ($\mathrm{WU}_{2025}/\mathrm{GDP}_{2025}$):

$$\boxed{\mathrm{GDP}_{2050}^{\text{fictitious}} = \mathrm{WU}_{2050} \times \frac{\mathrm{GDP}_{2025}}{\mathrm{WU}_{2025}} \approx 1,800\ \mathrm{W/cap} \times \frac{12.5\ \mathrm{kUSD/cap}}{508\ \mathrm{W/cap}} \approx 45\,\mathrm{kUSD/cap}}$$

This $\mathrm{GDP}_{2050}^{\text{fictitious}}$ is a measure of the **monetary inflation** necessary for GDP to appear to grow in lockstep with exergetic performance. However, the growth of WU is the result of **increased efficiency** and not an increase in spending. GDP becomes, in this scenario, an **accounting artifact** disconnected from physical reality.

7.5 Synthesis: GDP vs. WU

Criterion	Gross Domestic Product (GDP)	Useful Exergetic Well-being (WU)
Physical Unit	Monetary (USD)	Power (W)
Metrology	Expenditure / Transactions	Useful Exergy Flow $\dot{\Psi}_{\text{useful}}$
Nature	Nominal Flow / Debt	Real Flow / Sustainable
Vision	Unlimited Growth (Entropy)	Limited Efficiency (Negentropy)
Deflation	Counts Debt as Wealth	Reveals Debt as Ψ_{Mortgage}
Objective	Give Money to Humans	Give Energy to Humans

7.6 Conclusion: The Obsolescence of GDP

The thermodynamic model demonstrates that the goal of GDP maximization is fundamentally incompatible with sustainability and exergetic efficiency.
The increase in WU in the convergence scenario is not an increase in monetary wealth, but a **maximization of energy transformation efficiency** in the service of human well-being. GDP, designed for a world with high and growing EROI, becomes **obsolete** in a world constrained by thermodynamics.

$$\boxed{\mathrm{WU} \uparrow \text{(Efficiency)} \quad \Rightarrow \quad \mathrm{GDP} \downarrow \text{(Monetary Yield) and becomes an inflation artifact}}$$

8 Proposed Architecture: From Hyperbitcoinization to the Joule-Currency

Bitcoin is theorized in this work not as a final currency, but as a **thermodynamic transition mechanism** towards a universal unit of account. This transition is proposed in three distinct phases, the ultimate goal of which is the establishment of a **Joule-Currency (J-Currency)**, which realizes the ideal of the **Ideal Currency** (Nash [14]) anchored in physical laws (Georgescu-Roegen [25]).

8.1 Phase 1: Valuation of Curtailed Exergy (2025–2028)

8.1.1 Objective: Annihilation of the Carbon Emission Factor (CEF)

The initial objective is to neutralize the carbon impact of Bitcoin mining by exclusively capturing **surplus (curtailed) energy** from electrical grids. Mining then becomes a **network stabilization mechanism** that absorbs the excess production from renewables (solar, wind, hydro) without increasing net demand for fossil fuels.

8.1.2 Surplus Certificate Mechanism

To ensure that mining only consumes surplus exergy, a certification protocol is necessary:

- **Decentralized Installation**: Mining units are installed on or near renewable energy production sites.
- **Curtailment Oracle**: A network of decentralized oracles (e.g., based on Chainlink [26]) aggregates electrical grid load data (*Grid Data*).
- **Conditional Smart Contract**: The mining reward (Coinbase) is subject to the algorithmic condition that renewable production exceeds grid demand (a state of *curtailment*).

8.1.3 Impact Analysis

If $P_{\mathrm{BTC}} < P_{\mathrm{curtailment}}$, the network's **Carbon Emission Factor ($\mathbf{CEF_{BTC}}$)** tends toward zero by design:

$$\mathrm{CEF}_{\mathrm{BTC}} \to 0 \quad \text{if} \quad P_{\mathrm{BTC}} \subset P_{\mathrm{curtailment}}$$

In 2025, Bitcoin's annual consumption (≈ 80 TWh/year) represents about 32% of the estimated global surplus (≈ 250 TWh/year). The absorption of this surplus allows for the monetization of energy that would otherwise be **destroyed (entropy)**.

8.2 Phase 2: Exergetic Stabilization Mechanism for Bitcoin Price (SEP-BTC) (2028–2032)

8.2.1 Objective

To anchor Bitcoin's exchange value in a **measurable physical unit** to stabilize its volatility and prepare the transition to the J-Currency. Bitcoin's value is then officially correlated with the marginal cost of converted exergy.

8.2.2 Decentralized Exergetic Oracle

Implementation relies on decentralized aggregation of the physical price of energy.

Component	Technical Specification
Physical Sensors	Certified meters (e.g., IEC 62053-22 standard)
Oracle Network	Decentralized network of spot energy price observers
Aggregation	Weighted marginal price of the GJ on physical markets
Update Frequency	Synchronized with Bitcoin block time ($\approx$ 10 minutes)

8.2.3 Dynamic Peg Formula

The monetary price of Bitcoin (P_{BTC}) is then derived from a physical state function:

$$\boxed{P_{\text{BTC}} = \alpha \cdot P_{\text{GJ}} + \beta \cdot H + \gamma \cdot \phi_{\text{sat}}(t)}$$

with:

- P_{GJ} = **Marginal Price of the Gigajoule** (GJ) of exergy.
- H = **Hashrate** (network security, a function of expended energy).
- $\phi_{\text{sat}}(t)$ = Bitcoin's **Exergetic Penetration** (ratio between Ψ_{BTC} capitalization and global exergetic mass M_Ψ).
- α, β, γ = Calibration weights adjusted by protocol governance.

The current average backing of 1 BTC $\approx$ 19.8 MWh provides the asset with a **physical, universal, and measurable store of value** (Joules).

8.3 Phase 3: Architecture of the Exergy-Based Currency (J-Currency) (Post-2032)

The J-Currency is the final realization of a currency whose unit of account is the **Joule (J)** or Watt-second ($\text{W} \cdot \text{s}$), certified as having been transformed into **Useful Exergy** (Ψ_{useful}).

$$\boxed{1\,\text{J-Currency} = 1\,\text{W} \cdot \text{s of certified useful exergy}}$$

8.3.1 Proof-of-Exergetic Contribution (PoE) Consensus

The J-Currency consensus evolves beyond Bitcoin's PoW to become a PoE. In this model, the reward is allocated not for computation (which is a dissipation, albeit a useful one), but for the **certified production of useful exergy**.

$$\text{Reward}_i \propto \text{Stake}_i = \int_{t_0}^{t} \dot{\Psi}_{i,\text{useful}}(t)\, dt \quad [\text{J}] \tag{26}$$

- **Incentive**: PoE rewards the economic actor who **produces exergetic well-being** (e.g., care, education, efficient food production) and not the one who hoards debt.
- **Advantages**: Reduction of unnecessary energy dissipation, absence of monetary inflation through debt creation, and direct alignment of economic incentive with physical utility (WU).

8.3.2 Feasibility Case Study (MVP)

Component	Minimum Specification for a Local Pilot
Physical Sensor	Certified electrical / thermal meter
DLT Infrastructure	Exergetic Layer 2 (L2) or Sidechain
Validation (Oracle)	3 independent local nodes (Town Hall, University, Cooperative)
Issuance (Mining)	1 J-Currency is issued for 1 $W \cdot s$ injected into the network (or usefully consumed)

8.3.2.1 Example 1: Local Exergetic Microgrid In a rural community of 500 inhabitants powered by 50 kW of solar panels, annual production could generate $\approx$ 720 MJ/day (720×10^6 J-Currency/day). The unit of account becomes local: 1 J-Currency = 1 liter of purified water, ensuring that the currency is directly anchored in access to Essential Well-Being (WU_{min}).

8.3.2.2 Example 2: National Stabilization (Madagascar) A developing state could anchor 50% of its monetary mass (Ariary) to the J-Currency by 2030. The stability of the Ariary would be guaranteed by the certified national annual exergetic flow (≈ 5 TW $\cdot$ h/year), transforming the central bank into an **exergy production register** rather than a debt issuer.

8.4 Convergence Roadmap (2025–2050)

Period	Key Event	Modeled Impact (WU)
2025–2028	Phase 1: Mining on Surplus (VEC)	$CEF_{BTC} \to 0$
2028–2032	Phase 2: BTC Peg $\leftrightarrow$ GJ (SEP-BTC)	Stability and Revelation of exergetic cost
2032–2040	Phase 3: J-Currency Deployment (PoE)	WU increases through efficiency ($\alpha \uparrow 40\%$)
2050	J-Currency: Global Unit of Account	Convergence to $\mathbf{WU = 1,800}$ W/cap

Table 11: Theoretical roadmap for the thermodynamic transition, from Hyperbitcoinization to the Joule-Currency.

8.5 Conclusion: Bitcoin, the Bootstrapping Bridge

Bitcoin mining (PoW) is here conceptualized as the **thermodynamic bootstrapping mechanism**: it forces the monetization of surplus energy (Phase 1) and reveals the fundamental value of exergy (Phase 2).

> *"The PoW stage of Bitcoin is not the end goal, but the indispensable* ***transitional bridge*** *to shift from a currency backed by debt and entropy to a* ***physical currency (J-Currency)*** *whose value is defined by transformed useful energy."*

The optimal allocation of resources, necessary to manage global energy constraints (low $EROI_{net}$), requires an irrefutable unit of account. The J-Currency provides this unit.

$$\boxed{\text{Bitcoin (PoW)} \xrightarrow{\text{Transition Mechanism}} \text{J-Currency (PoE)} \quad \Rightarrow \quad \text{Value} \propto \Delta\Psi_{\text{useful}}}$$

9 Conclusion: Thermodynamic Convergence and the Imperative for an Exergy-Anchored Currency

This research formally establishes a unified theoretical framework for value, societal well-being, and monetary architecture, fundamentally grounded in the principles of non-equilibrium thermodynamics and economic biophysics. Our primary contributions are structured around three core axes:

1. **Metrology of Well-Being and Efficiency**: We introduce the concept of **Exergetic Well-being per capita (WU)**, a rigorous physical metric proposed as a successor to GDP. WU intrinsically integrates utility, equity, and resource sustainability, defined by the relationship $\mathrm{WU} = \frac{\dot{\Psi}_{\text{useful}}}{P} \times f(\dots)$. The theoretical foundation draws inspiration from the concept of *Wu Wei*, utilizing Exergy as the core physical factor to establish a non-monetary proxy for societal prosperity. Our analysis confirms a pronounced and increasing **exergetic inefficiency** within advanced monetary economies.

2. **Revelation of Systemic Entropic Risk**: The formalized extension of the Kaya identity to the **Kaya-Ψ-WU** equation (Section 6) demonstrates that the current monetary architecture of debt intrinsically generates a substantial **carbon mortgage**. Calibration of the exergetic amplification factor ($\lambda \simeq 3.5$) reveals that **78**% of global CO_2 emissions are structurally attributable to the entropic maintenance costs of this unsustainable debt paradigm.

3. **Proposed Exergetic Monetary Architecture**: We model the Bitcoin Proof-of-Work (PoW) consensus as an **entropic transition mechanism** towards the **Joule-Currency (J-Currency)**, where the definitive unit of account is the Joule of certified useful exergy (Section 8).

9.1 The Bitcoin Protocol and the Imperative of the Fourth Law of Thermodynamics

The concept of an ideal currency, as articulated by John Nash [14, 15], which requires asymptotic stability, finds its ultimate physical realization in energy. Specifically, transformed useful exergy satisfies the five fundamental criteria of any perennial unit of account: universality, direct utility, structural resistance to inflation, physical scalability, and systemic equity.
The Bitcoin Proof-of-Work (PoW) Protocol serves as the **first coded expression of the Fourth Law of Thermodynamics** (Georgescu-Roegen [25]) within the monetary domain:

- **Physical Scarcity**: The terminal supply of 21 million BTC is the monetary analogue of the **irreversible, finite limitation of the Earth's low-entropy stock**.

- **Entropic Anchoring**: The mining process (PoW) is an irreversible energy expenditure that anchors monetary value in a **measurable physical transformation**, thereby structurally eliminating *ex nihilo* monetary issuance.

This dual constraint reveals the fundamental incoherence between the objective of maximizing GDP (a goal of unlimited growth) and the imperative of maximizing WU (a goal constrained by physical efficiency).

Thermodynamic Axiom		Monetary Consequence
Low-entropy stock is finite (4^{th} Law)	$\Longrightarrow$	Capped monetary supply (**21** M BTC)
All value is exergy (Odum)	$\Longrightarrow$	Unit of account $\propto \Delta\Psi_{\text{useful}}$ (**J − Currency**)
$\lambda \gg 1$ (Unsustainable Debt)	$\Longrightarrow$	**Forced decarbonization** by monetary purge

Table 12: Synthesis of the convergence between governing physical laws and the monetary architecture of the future.

9.2 Implications and Future Perspectives

The proposed J-Currency (based on Proof-of-Exergy, PoE) is not merely an alternative currency, but an **Exergetic Sufficiency Protocol**. It empowers communities to align their local economy with planetary thermodynamic limits by rewarding energy produced and utilized in a demonstrably useful and sustainable manner.

The convergence toward an exergy-based unit of account will permit Artificial Intelligence (AI), through its power of optimization, to become the **first scientific tool capable of optimally allocating resources** for the maximization of Universal Exergetic Well-Being (WU).

The critical challenge is no longer technological, but fundamentally paradigmatic: accepting the constraints of **physical scarcity** to achieve the goal of **human sufficiency**.

$$\boxed{\text{GDP} \to \mathbf{WU} \quad \text{and} \quad \text{Debt} \to \boldsymbol{\Delta\Psi}_{\text{useful}}}$$

The future of money will be peaceful because it will be **scientific**: constrained by the only law that humanity cannot print—the conservation of energy.

> **Foundational Proposition:** *"If we were compelled to restart civilization with a single unit of account, it would be energy, measured in Joules. Nothing else can lay claim to universality, intrinsic utility, and systemic resistance to inflation."*

References

[1] Ilya Prigogine and Isabelle Stengers. *La Nouvelle Alliance : Métamorphose de la science*. Gallimard, 1979. Édition originale : Order Out of Chaos, 1977.

[2] Food and Agriculture Organization of the United Nations. The state of food and agriculture 2025. Technical report, FAO, July 2025. Consommation énergétique humaine : 2 000 kcal/jour = 100 W.

[3] Bitmain Technologies. Antminer s21 product specification sheet, 2025. Puissance : 3 500 W ; Hashrate : 200 TH/s ; Efficacité : 17,5 J/TH.

[4] Jan Szargut. *Exergy Analysis of Thermal, Chemical, and Metallurgical Processes*. Hemisphere Publishing Corporation, 2005.

[5] Weston A. Hermann. Quantifying global exergy resources. *Energy & Environmental Science*, 13(8):2211–2220, 2020.

[6] Robert U. Ayres and Benjamin Warr. *The Economic Growth Engine: How Energy and Work Drive Material Prosperity*. Edward Elgar Publishing, 2010. Corrélation exergie $\leftrightarrow$ PIB.

[7] International Energy Agency. World energy outlook 2025. Technical report, IEA, 2025.

[8] Jonathan M. Cullen and Julian M. Allwood. The efficient use of energy: Tracing the global flow from fuel to service. *Energy Policy*, 38(1):75–81, 2010. Cascade exergétique mondiale.

[9] Jonathan M. Cullen and Julian M. Allwood. The efficient use of energy: Tracing the global flow from fuel to service. *Energy Policy*, 38(1):75–81, 2010.

[10] Robert U. Ayres and Benjamin Warr. *The Economic Growth Engine: How Energy and Work Drive Material Prosperity*. Edward Elgar Publishing, 2009.

[11] International Energy Agency (IEA). Electricity consumption and efficiency trends in major economies (2023 data), 2024. Consulté le 20 novembre 2025. Données utilisées pour la conversion en GW/an.

[12] Cambridge Centre for Alternative Finance (CCAF). Cambridge bitcoin electricity consumption index (cbeci), 2023. Données de consommation de Bitcoin (GW). La valeur de 120 GW est une extrapolation du seuil critique $\phi = 1\%$.

[13] Pascal Ranaora. Bitcoin et l'Économie de l'Énergie: Une théorie unifiée du bien-être exergétique et de la monnaie universelle en joules. *Pre-print: Bitcoin Thermodynamics*, 2025. Source interne pour le seuil critique $\phi = 1\% \Rightarrow 120$ GW.

[14] John F. Nash. Ideal money. Southern Economic Journal, Vol. 69, No. 1, July 2002. Conférence prononcée en 2002.

[15] John F. Nash. *Ideal Money and Asymptotically Ideal Money*. Independently published, 2017. Compilation posthume des travaux sur la monnaie idéale.

[16] Charles A. S. Hall and Kent A. Klitgaard. *Energy and the Wealth of Nations: Understanding the Biophysical Economy*. Springer, 2011.

[17] Yoichi Kaya. Impact of carbon dioxide emission control on gnp growth: Interpretation of proposed scenarios. *Paper presented to the Intergovernmental Panel on Climate Change (IPCC) Energy and Industry Subgroup*, 1990. Original formulation of the Kaya Identity.

[18] BP. Statistical review of world energy 2025. Technical report, BP p.l.c., 2025.

[19] United Nations Development Programme. Human development report 2025. Technical report, UNDP, 2025.

[20] World Bank. World development indicators 2025. Technical report, World Bank, 2025.

[21] Global Footprint Network. Ecological footprint explorer 2025. Technical report, GFN, 2025.

[22] Howard T. Odum. *Environment, Power, and Society.* Wiley-Interscience, New York, 1971.

[23] Howard T. Odum and Elisabeth C. Odum. *A Prosperous Way Down: Principles and Policies.* University Press of Colorado, 2001.

[24] François Roddier. *Thermodynamique de l'évolution : Un essai de thermo-bio-sociologie.* Parole d'Aube, Vaux-en-Velin, 2012.

[25] Nicholas Georgescu-Roegen. *The Entropy Law and the Economic Process.* Harvard University Press, 1971.

[26] Sergey Ellis, Ari Griese, Andrew Kotsyba, Samuel Odeniran, and Ben Van de Sande. Chainlink 2.0: Next Steps in the Evolution of Decentralized Oracle Networks. 2021. Whitepaper/Document technique (version 2.0).